

Bounded cluster avoidance for uncountable selector families

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A complete written proof with an independent model-conducted mathematical audit. No human peer review, proof-assistant formalization, or novelty claim is implied.

Result and relation to the existing entries

This note proves an uncountable-family extension of the routing argument in mxym/math entries 004 and 006. Instead of making one candidate at a scale hit a small open set, it makes an entire bounded finite cluster hit. Every pointwise selector from those clusters is then excluded by the same large closed set. The proof does not enumerate selectors or approximate them by a countable family of leading profiles.

Entry 003 concerns a different invariant: upper box dimension zero and Assouad dimension two of a Banach-space nonembedding obstruction. Its covering proof suggests keeping branching cardinality separate from scale count, but its nonembedding theorem is not an analytic dependency here. In the example below the *space of profiles* has Assouad and upper box dimension one, whereas every individual null configuration has upper box dimension zero. Neither of these statements identifies those dimensions with the logarithmic upper Banach density used in 004. The critical-covering refinement of 003 is recorded separately in [the critical-covering note](#).

Theorem A

Let S be a subset of the positive integers with $d^*(S) > 0$. For every j in S let F_j be a nonempty finite set satisfying

$$F_j \subset [2^{-j-1}, 2^{-j}], \quad |F_j| \leq K.$$

where K is a fixed positive integer. Let (ω_r) be a nonempty countable family of nondecreasing finite null moduli, meaning $\omega_r(a)$ tends to zero as a tends to zero.

For every $0 < \epsilon < 1$ there is a closed nowhere-dense one-periodic set E such that $|E \cap [x, x+1]| > 1 - \epsilon$ for every real x and the following holds. For every r, y in $\mathbb{R}$, $c \neq 0$, M finite and nonnegative, and every collection of errors $e_{j,a}$ satisfying

$$|e_{j,a}| \leq M a \omega_r(a).$$

on all sufficiently late clusters, there are arbitrarily large j in S for which

$$\{y + ca + e_{j,a} : a \in F_j\} \subset \mathbb{R} \setminus E. \quad (\text{A.1})$$

Consequently, every pointwise selector a_j in F_j and every perturbed sequence $v_j = y + ca_j + e_j$ with $|e_j| \leq Ma_j\omega_r(a_j)$ has infinitely many distinct values outside E on every tail.

The same E works for any prescribed countable family of such cluster systems; each system may have its own finite bound K and positive density.

The closed annulus convention is harmless. It permits the useful example $F_j = \{2^{-j-1}, (3/2)2^{-j-1}\}$ directly. There is no assertion for unbounded cluster cardinalities or for a continuous interval of candidates at each scale. The null moduli remain prescribed and countable.

Conventions

Here $d^*(S)$ is upper Banach density: the limit of $F_S(L)/L$ defined below. Measure is Lebesgue measure. A configuration has positive logarithmic upper Banach density when the set of its occupied dyadic bins has positive upper Banach density. A log-bi-Lipschitz null profile is a strictly increasing continuous function ϕ on a small interval $(0, a_0]$, tending to zero at zero, for which $\Psi(z) = -\log_2 \phi(2^{-z})$ satisfies $\alpha(v - u) \leq \Psi(v) - \Psi(u) \leq \beta(v - u)$ for all sufficiently large $u < v$, with fixed $0 < \alpha \leq \beta < \infty$. Each prescribed profile may have its own constants. This is the convention of entry 006 v2.

Density template placement

We recall and prove precisely the input from 004/006. Put $F_S(L) = \sup_m |S \cap \{m + 1, \dots, m + L\}|$ and $\delta = d^*(S)$. The limit $F_S(L)/L$ exists by subadditivity and equals its infimum. For every $\xi > 0$ there is a constant C_ξ such that every integer interval I obeys

$$|S \cap I| \leq (\delta + \xi)|I| + C_\xi. \quad (1)$$

Indeed choose L_0 with $F_S(L_0) \leq (\delta + \xi)L_0$ and partition I into blocks of length L_0 and a remainder; $C_\xi = L_0$ suffices. Every tail of S has the same upper Banach density. For every cutoff H and every length T , some interval of length T beyond H contains at least δT points. Choose an interval maximizing the count for that translated tail, rather than requiring it to maximize the original S . The supremum is attained since the possible counts are integers in a finite set.

Fix $0 < \eta < \delta$. Suppose an interval template has total span $T_L \leq C_0 L$ and contains finitely many windows, each of length at least L . Put the template on a length- T_L interval J beyond any requested initial index that contains at least δT_L points, as supplied by the translated-tail argument. For every window W , its complement in J consists of at most two intervals. Applying (1) there gives

$$|S \cap W| \geq \delta|W| - \xi C_0 L - 2C_\xi.$$

Choose $\xi C_0 < (\delta - \eta)/2$ and then L sufficiently large. Every window contains at least $\eta|W|$ selected indices. The threshold for L is independent of the requested initial index. This independence is needed for buffering.

One parity of S has positive upper Banach density; otherwise subadditivity would force $d^*(S)=0$. Replace S by that parity. Blocking these clusters is enough for Theorem A, and discarding any finite initial segment is harmless. Hereafter δ and η refer to this chosen parity.

Finite routing with entire clusters

Fix $0 < p < 1/12$ and define

$$\theta = (p/2)^K.$$

Choose an integer $M \geq 2$ with $\theta\eta(M-1) > 4$. Choose a depth d with $(1 - 2^{1-M})^d < p$. Let Q be the number of edges in a complete ordered M -ary rooted tree of depth d . Choose an integer gap $g \geq 1$ with $Q2^{3-g} < p$.

Order edges in preorder. Edges leaving a vertex of height h have common window length ℓ_h . Put g unused integer indices between consecutive windows. For a base length $L \geq g$ set

$$\begin{aligned} \ell_1 &= L, & \sigma_1 &= ML + (M-1)g, \\ \ell_h &= g + \sigma_{h-1}, \\ \sigma_h &= M(\ell_h + g + \sigma_{h-1}) + (M-1)g \quad (h \geq 2). \end{aligned}$$

The total span $T = \sigma_d$ is at most $C_0 L$ for fixed M, d, g . If an edge window is $[u_e, v_e]$ and v_e^* is the final endpoint of it and its child subtree, then $v_e^* - u_e + 1 \leq 2\ell_h$. Apply the density template lemma to place it beyond any initial index, with at least $\eta\ell_h$ selected indices per window. Require the initial index to be at least four.

Use one-periodic nested grids with $N_b = 2^{b+3}$ cells and key $J_b(z) = \lfloor N_b\{z\} \rfloor$. For every nondefault edge $e = (P, i)$, $i < M$, assign an independent fair-bit selector table at grid v_e . The M th child is default. Every leaf has an independent Bernoulli- p terminal table at its incoming edge's grid. All entries in all tables are independent.

Route z to the first nondefault child whose addressed selector is one, or to the default when all are zero, and continue to a leaf. Let B be the set of z whose addressed terminal bit is one. Conditioning on all selectors shows $E \text{ density}(B) = p$. Every outcome is constant on a finest periodic grid with $N = 2^{u+T+2}$, where u is the template start.

A center x is stable if for each predecessor endpoint b' and next window start $u' = b' + g + 1$, the interval $(x, x + 2^{-(1-u')}]$ contains no point of the grid $N_{b'}(-1)$. The unstable centers have density at most $Q2^{3-g} < p$. For a stable x and any candidate a in F_j with j in a window, every $x + t a$, $1 \leq t \leq 2$, agrees with x at all grids of earlier edges. This follows from $t a \leq 2^{-(1-u_e)}$, the stability condition and nesting.

Candidates from *different* selected indices in one window have distinct keys at that window's grid, and avoid the center key. To check this, if $j = j+2$, then

$$\min F_j - \max F_k \geq 2^{-j-1} - 2^{-k} \geq 2^{-j-2} \geq 2^{-v_e-2}.$$

which is twice the cell width. Also every candidate is at distance at least $2^{-(v_e-1)}$ from the center. All displacements lie in an interval of length at most $1/8$, so circular wraparound creates no collision.

The separation persists at every finer grid. Candidates *within* one cluster may share keys, and we allow this explicitly.

Fix a stable x and expose its addressed selector entry in every selector table, but expose no terminal entries. This determines x 's route. Its probability of having no default is $(1 - 2^{1-M})^d < p$. On an exposure atom whose first default is at U of height h , test every selected j in each of U 's first $M-1$ outgoing windows. For an edge $e_i=(U,i)$, route each $z=x+t$ locally from its child U_i . If every own-edge selector at that cluster is one and every resulting local terminal entry is one, the entire cluster $x+t$ belongs to B . Earlier routing decisions agree with x ; therefore all its candidates really enter U_i .

The new probability estimate

There are $m \geq \eta(M-1)\ell_h$ tested clusters. For each cluster, let A_j be the event that all its own-edge selector entries are one. It consults at most K distinct entries, so $\mathbb{P}(A_j \mid \text{center exposure}) \geq 2^{-K}$. Different clusters consult disjoint own-edge entries, by the separation just proved (or use different tables). Thus their A_j events are independent under the center exposure.

Now condition on *all* selectors. Let d_j be the number of distinct local terminal addresses within cluster j ; $1 \leq d_j \leq K$. Addresses belonging to different clusters are distinct: different children have disjoint subtrees; different leaves have different tables; at the same leaf, different index-clusters retain distinct keys at the finer grid. Conditional on the selectors, the terminal-success events for different clusters are therefore independent, and the success probability of cluster j on A_j is $\hat{p}(d_j)$. The conditional probability that all tested clusters fail is

$$\prod_j [1 - \mathbf{1}_{A_j} p^{d_j}] \leq \prod_j [1 - \mathbf{1}_{A_j} p^K].$$

Averaging over selectors and using the independent A_j events gives

$$\mathbb{P}(\text{all clusters fail} \mid \text{center exposure}) \leq (1 - (p/2)^K)^m \leq e^{-\theta m}. \quad (2)$$

This estimate is an inequality, not the singleton engine's exact identity. Collisions within a cluster can only improve its success probability. Shared auxiliary selectors do not invalidate the calculation: they were conditioned on first, and d_j was bounded before averaging.

Uniformity over every normalized dilation

A candidate in a tested cluster on edge e_i consults only grids up to $v_{e_i}^*$. As t traverses $[1,2]$, its entire local outcome changes at at most $1 + 2^{2\ell_h+3}$ deterministic boundary values. There are at most $K(M-1)\ell_h$ candidates. Represent every boundary, both endpoints, and one point of every intervening open interval. At most

$$D_{K,M}(\ell) = 4 + 2K(M-1)\ell(1 + 2^{2\ell+3}).$$

representatives suffice. The whole test vector is constant between the represented boundaries for every table outcome. Thus the probability that some t has no entire successful cluster is at most

$$D_{K,M}(\ell) \exp[-\theta\eta(M-1)\ell]. \quad (3)$$

Since $\theta\eta(M-1) > 4 > 2\log 2$, choose L large enough that (3) is less than p for every integer $\ell \geq L$. The parameters M, d, g, L, T are all chosen before the actual template location. Combining the no-default cost with (3) gives probability less than $2p$ at every stable center. Boundaries are represented individually; failures supported only at a boundary are retained.

Robust normalized blocker

Given a null modulus ω , let T be the finite span just constructed. For a possible start u put

$$\eta_u = \omega(2^{-u}) + 2^{-u}, \quad r_u = 2^{-u}\eta_u > 0.$$

Choose a lower bound on u so late that

$$4Nr_u = 2^{T+4}\eta_u < p, \quad N = 2^{u+T+2}.$$

This holds at every later start. The density-template lemma still supplies a filled template there. Every test candidate $a \leq 2^{-u}$ has $|e_{\{j,a\}}| \leq \omega(a) \leq r_u$.

For each random B let $B_1 = B + (-r_u, r_u)$ and $B_2 = B + (-2r_u, 2r_u)$ on the circle. If B has N grid cells, enlarging them gives

$$\text{dens}(B_2) \leq \text{dens}(B) + 4Nr_u.$$

so $\mathbb{E} \text{dens}(B_2) < 2p$. Let $I = S$ intersect the union of all edge windows be the deterministic finite test-index set. Define the exceptional-center set

$$R = \{x : \exists t \in [1, 2] \forall j \in I, \\ \exists a \in F_j \text{ with } x + ta \notin B_1\}.$$

It is closed and periodic: the failure condition is a finite intersection of finite unions of closed conditions on the compact rectangle. Its projection onto the circle is closed. By the finite estimate and unstable center cost, $\mathbb{E} \text{dens}(R) \leq 3p$. Choose a table outcome with

$$\text{dens}(B_2) + \text{dens}(R) < 5p.$$

By outer regularity choose an open periodic V containing R with $\text{dens}(V) < \text{dens}(R) + p$. Put $H = B_2 \cup V$; $\text{dens}(H) < 6p$.

If x is not in R , one tested cluster lies entirely in B_1 . Any collection of errors of size at most r_u puts the entire cluster in B_2 : the first buffer gives a strict margin and the second absorbs endpoint equality. If x is in R , it lies in V . As j tends to infinity in S , every candidate in F_j and every allowed error tends uniformly to zero. Therefore all sufficiently late perturbed clusters lie in V . This repairs every center, not merely almost every center.

We have proved: for every x , every t in $[1,2]$, and every allowed independent error array, at least one entire cluster $x+tF_j+\text{errors}$ lies in H . The successful index for a fixed x,t can be chosen before the error array: the whole closed uncertainty interval for every candidate is contained in H . The statement may be applied to any specified tail and any dyadic dilation of the cluster system.

All coefficients and infinitely many misses

For every r , integer k , integer $q \geq 1$ and tail cutoff h , apply the normalized blocker to clusters $2^k F_j$ for $j \geq h$, deleting any finitely many nonpositive annular indices. Their annular indices are shifted by $-k$ and retain positive upper Banach density. Use the transformed null modulus

$$\Omega_{r,k,q}(b) = q2^{-k}\omega_r(2^{-k}b).$$

Choose positive budgets $p_{-}(r,k,q,h) < 1/12$ with $12 \sum_{r,k,q,h} p_{r,k,q,h} < \epsilon$. Let U be the union of all blockers and their reflections, and $E = \mathbb{R} \setminus U$. This is closed and one-periodic, and its complement has density less than ϵ .

For $c > 0$ write $c = 2^k t$ with $1 \leq t \leq 2$, and choose q at least the finite error constant M from Theorem A (not the local routing arity). At $b = 2^k a$ the error bound $M a \omega_r(a)$ is at most $\Omega_{-}(r,k,q)(b)$. For every sufficiently late h , the component with that tail supplies an entire cluster as in (A.1). Thus the good cluster indices are unbounded. For $c < 0$ use the reflected blocker and negate the errors.

Every selected v_j obeys, on sufficiently late clusters,

$$(|c|/2)a_j \leq |v_j - y| \leq (3|c|/2)a_j.$$

Consequently these outside values tend to y but never equal it there, which guarantees infinitely many distinct outside values on every tail. Countably many cluster systems can be added as an extra index to the summable budget.

Finally E has empty interior. Choose once and for all one selector a_j from a bounded tail of the first cluster system. If an interval lay in E , a sufficiently small translated affine copy of this bounded selector would fit in that interval. The already proved assertion with zero errors forbids this. A closed set with empty interior is nowhere dense. Theorem A is proved.

An explicit uncountable log-bi-Lipschitz family

Let $A = \{2^{-n} : n \geq 1\}$, and for every binary sequence σ with $\sigma_n \in \{1, 3/2\}$, prescribe

$$\phi_{\sigma}(2^{-n}) = \sigma_n 2^{-n}.$$

Define Ψ_{σ} on $[1, \text{infinity})$ by linear interpolation of $\Psi_{\sigma}(n) = n - \log_2 \sigma_n$, and set $\phi_{\sigma}(a) = 2^{-\Psi_{\sigma}(-\log_2 a)}$ for $0 < a \leq 1/2$, with $\phi_{\sigma}(0) = 0$. Every segment slope belongs to

$$[1 - \log_2(3/2), 1 + \log_2(3/2)].$$

Both endpoints are positive. Hence every ϕ_σ is a strictly increasing log-bi-Lipschitz null profile with these common constants. Theorem A applied to $F_{n-1} = \{2^{-n}, (3/2)2^{-n}\}$, $n \geq 2$, produces one E excluding every image $y+c \phi_\sigma(A)$, simultaneously for every σ , every $c \neq 0$, and every y . It also permits the displayed prescribed null perturbations.

This is an uncountable upgrade: σ may change independently at every scale. A selector with infinitely many changes has no fixed constant multiple of the identity as its leading profile. More generally, a countable family of arbitrary fixed profiles cannot cover all these selectors by relative $o(1)$ approximation, even allowing separate scalar coefficients. If two selectors were asymptotic to scalar multiples of the same profile, their quotient on A would converge. That quotient belongs to $\{2/3, 1, 3/2\}$, so it is eventually constant. Apart from the two constant-tail classes, this means the two selectors eventually agree. Each eventual- equality class is countable, while there are continuum many such classes. This is why countable union bookkeeping in 006 alone does not supply the new simultaneous quantifiers.

Equip the binary parameter space with $d(\sigma, \tau) = 2^{-m}$, where m is the first differing index. It has Assouad dimension and upper box dimension exactly one, by counting binary cylinders. Under the uniform function norm on $[0, 1/2]$, the map $\sigma \rightarrow \phi_\sigma$ is bi-Lipschitz:

$$(1/2)2^{-m} \leq \|\phi_\sigma - \phi_\tau\|_\infty \leq 3 \cdot 2^{-m}.$$

The lower bound is the difference at $a=2^{-m}$. Before $z=m-1$ the profiles agree; afterwards both have size at most $3 \cdot 2^{-m}$, which gives the upper bound. Thus the profile family itself has both dimensions one. Its individual image sequences have upper box dimension zero, since they decrease geometrically. The density of their logarithmic bins is positive. Those are three separate statements about three separate objects.

These profiles are also uniformly bi-Lipschitz in the ordinary variable away from zero, and extend continuously to zero; on an interpolation segment interior their derivatives are the logarithmic slope times $\phi_\sigma(a)/a$. The ratio $\phi_\sigma(a)/a$ lies in $[1, 3/2]$, so the derivative lies between $1 - \log_2(3/2)$ and $(3/2)(1 + \log_2(3/2))$. These common bounds imply ordinary bi-Lipschitz estimates, which extend to zero by continuity. This does not assert all such profiles are C1 at zero. For non-eventually-constant σ , $\phi_\sigma(2^{-n})/2^{-n}$ does not converge, so differentiability at zero already fails.

The theorem therefore does not contradict the known positive embedding results for arbitrary bi-Lipschitz maps or the C1 endpoint in entry 006. It excludes this explicit finite-alphabet family, not all maps with the same distortion constants, all real power exponents, or flat germs.

Exact finite certificates

A certificate lists finitely many rational clusters, one nonnegative rational error radius r_a per candidate, finitely many rational OPEN intervals repeated with period one, and a rational density budget. It asserts

$$\begin{aligned} \forall x \in [0, 1] \ \forall t \in [1, 2] \ \exists \text{ a listed cluster } F : \\ [x + ta - r_a, x + ta + r_a] \subset H \quad (\forall a \in F). \end{aligned} \tag{C.1}$$

Failure means choosing one failed candidate from every cluster, and then a closed complementary gap $[l, u]$ reachable by an allowed error. For each such finite choice the constraints are

$$0 \leq x \leq 1, \quad 1 \leq t \leq 2, \quad l - r_a \leq x + ta \leq u + r_a.$$

Exact half-plane clipping in two rational variables decides feasibility. Segments and singletons are retained, so endpoint-only failures remain. If feasible, the checker chooses a rational x, t and independently verifies one legal escaping error in every cluster. If no choice is feasible, (C.1) holds. The density is computed by merging genuinely overlapping open intervals; merely touching intervals retain a singleton gap.

The implementation `check_cluster_cover.py` uses Fraction arithmetic only. Seven regression cases pass. The supplied computed certificate is

$$\begin{aligned} F_n &= \{2^{-n}, (3/2)2^{-n}\}, \quad n = 1, 2, 3, 4; \\ r_a &= a/100; \\ H &= (0, 4/5) + \mathbb{Z}, \quad \text{density budget } 4/5. \end{aligned}$$

It returns valid after nine feasible-polygon states. The first three clusters alone fail, and an exact counterexample is $x = 7/15$, $t = 1591/900$, with legal escaping candidates $3/4$, $1/4$ and $3/16$. The four-cluster certificate is nontrivial and tests the stronger all-candidates assertion. It is not a computed small-density witness for every ϵ .

Under the bounded-cardinality and positive-annular-density hypotheses of Theorem A, for computably listed rational clusters and computable rational radii with r_a/a uniformly tending to zero by cluster index, there is also a terminating exact search for normalized certificates of any rational positive density budget. The relative radii admit a finite nondecreasing null-modulus majorant on the annular tail, obtained by taking the supremum of r_a/a over all candidates below a given scale. No computable convergence rate is needed for existence. The robust normalized proof therefore provides an open H of strictly smaller density. Entire closed uncertainty intervals are contained in H for a successful cluster, and this is an open condition on x, t . Compactness supplies a finite subcover by successful clusters. Shrink to finitely many rational open intervals inside H still covering those compact uncertainty intervals, and choose rational endpoints so the strict density budget survives. Enumerate finite rational interval lists and cluster prefixes and run the exact verifier. Existence proves the search terminates. No feasible runtime or complexity bound is claimed.

Primary source comparison and scope

1. OpenAI, The geometric case of the Erdos similarity conjecture, family 084, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its theorem is for each fixed geometric ratio. Sections 04-routing and 05-scales are the source of center exposure, conditioning on all selectors, finite dilation entropy, and exceptional-center repair. The bounded-cluster probability estimate (2) is the added step here. [Pinned routing source](#) [Pinned scales source](#)
2. mxym/math entry 004 v1.1 supplies the positive-logarithmic-upper-Banach- density template extension; entry 006 v1 supplies translation-uniform placement and the two-buffer error argument; entry 006 v2 treats a prescribed countable family of log-bi-Lipschitz profiles. The exact source snapshots are bundled with this note and bound to the immutable mxym/math commit `1c67b4146f9e54291c8a5ca1f42a028bf307629f`; 006 v2 explicitly disclaims an arbitrary uncountable profile family. [004 v1.1 source](#) [006 v1 source](#) [006 v2 source](#)

3. Feng, Lai, Xiong, Erdos similarity problem via bi-Lipschitz embedding, arXiv:2312.01319v1, Theorem 1.1: a geometrically decaying null sequence has a bi-Lipschitz image in every positive-measure set, even with derivative one at zero. This is consistent with the restricted family here, and warns against claiming all bi-Lipschitz or all C1 images. [Primary source](#)
4. Bellissard and Julien, Bi-Lipschitz Embedding of Ultrametric Cantor Sets into Euclidean Spaces, arXiv:1202.4330v2, gives geometric context for counting weighted binary-tree cylinders and finite Assouad dimension. The elementary dimension calculation above is proved here directly. [Primary source](#)

This is a written proof with exact finite regression checks and an independent model-conducted mathematical audit. It is not human peer review, formal verification, or a literature-priority assertion. The main argument has no numerical optimization dependency. See [the proof audit](#), [verification instructions](#), and [the source provenance](#).

Extension through the countable log-bi-Lipschitz profiles of 006 v2

The theorem also applies to clusters in a fixed-width logarithmic annulus. More precisely, replace its annulus condition by

$$F_j \subset [\lambda 2^{-j}, \Lambda 2^{-j}], \quad 0 < \lambda \leq \Lambda < \infty.$$

with λ and Λ fixed for the cluster system. First divide all candidates by Λ . Choose an integer $q \geq 1$ with $2^{-q} \leq \lambda/\Lambda$. One residue class of S modulo $q+1$ has positive upper Banach density; keep it. Use grids $N_b = 2^{b+q+3}$, rather than 2^{b+3} . Distinct selected index clusters are separated by at least 2^{-j-q-1} ; this is at least four grid cells at an edge endpoint $b \geq j$, and each candidate avoids the center key. All other arguments are unchanged with the following explicit replacements:

$$\begin{aligned} \text{unstable density: } & Q2^{q+3-g}; \\ \text{entropy: } & 4 + 2K(M-1)\ell(1 + 2^{2\ell+q+3}); \\ \text{finest grid: } & N = 2^{u+T+q+2}; \\ \text{buffer cost: } & 4Nr_u = 2^{T+q+4}\eta_u. \end{aligned}$$

Choose g so $Q2^{q+3-g} < p$. The entropy exponential rate is still $2 \log 2$, so $\theta\eta(M-1) > 4$ suffices. Positive λ and finite q keep all choices finite; no dependence on the eventual translation or tail location is introduced. The preliminary division by Λ is absorbed into the free coefficient and into the pulled-back null modulus. This proves the wide-annulus version of Theorem A with exactly the same conclusion.

Corollary B

Let (A_l) be a nonempty countable family of configurations with positive logarithmic upper Banach density. Let (ϕ_r) be a prescribed nonempty countable family of log-bi-Lipschitz null profiles as in 006 v2. Let (D_v) be a nonempty countable family of finite nonempty subsets of $(0, \infty)$, and let (ω_w) be a prescribed nonempty countable family of null moduli. For every $\epsilon > 0$ one closed nowhere-dense one-periodic set E of measure more than $1-\epsilon$ in every unit interval has the following property.

For every choice of l, r, v, w, y in $\mathbb{R}$, $c \neq 0$ and M finite, every function f on a sufficiently small tail of A_1 satisfying

$$\begin{aligned} f(a) &= y + cd(a)\phi_r(a) + e(a), \\ d(a) &\in D_v, \\ |e(a)| &\leq M\phi_r(a)\omega_w(a). \end{aligned} \tag{B.1}$$

has infinitely many distinct values outside E on every sufficiently small tail. The multiplier $d(a)$ may switch arbitrarily at every input scale; no regularity or enumeration of these switching functions is assumed.

Proof. By 006 v2 Lemma 2.1, $B = \phi_r(A_1)$ has positive logarithmic upper Banach density. Choose one $b_j = \phi_r(a_j)$ from every occupied dyadic bin, then use the cluster D_v b_j . Its cardinality is at most $|D_v|$, and it lies in $[\min(D_v)2^{-j-1}, \max(D_v)2^{-j}]$, so the wide-annulus theorem applies. Pull ω_w back to the b -coordinate by $\Omega(b) = \omega_w(\phi_r^{-1}(b))$, extending constantly beyond a small interval when necessary. It is a null modulus. Put $d_0 = \min D_v > 0$ and define the candidate-coordinate null modulus $\Omega_{\text{new}}(u) = \Omega(u/d_0)$. Since $d \geq d_0$, an error bounded by $Mb\Omega(b)$ is at most $(M/d_0)(db)\Omega_{\text{new}}(db)$, exactly the per-candidate bound required by the cluster theorem. Prescribe this same error at every candidate d b in a cluster; whole-cluster success then covers the actual choice $d(a_j)$.

The countable indices l, r, v, w are absorbed into the summable density budget. Unbounded successful cluster indices imply $b_j \rightarrow 0$, hence $a_j \rightarrow 0$ because ϕ_r is increasing and tends to zero. On sufficiently small inputs, $|f(a) - y|$ is between positive constant multiples of $\phi_r(a)$, since D_v is finite and bounded away from zero and $\omega_w(a) \rightarrow 0$. Thus the outside hit values tend to y without equalling y , giving infinitely many distinct values on every tail. This proves the corollary.

Taking $D = \{1\}$ recovers the relevant profile statement of 006. Taking $D = \{1, 3/2\}$ permits genuinely arbitrary switching with no fixed leading profile and gives the stronger uncountable simultaneous conclusion. Countability still applies to the prescribed profiles, alphabets and error moduli. No simultaneous claim for all positive real exponents follows.

Additional finite check: using the same candidate formula and relative error radius $a/100$, the first six clusters certify $H = (0, 3/4) + \mathbb{Z}$, with exact density $3/4$, in fourteen polygon states. This is saved separately as `verification/toy_cluster_certificate_3_4.json` and `verification/toy_cluster_result_3_4.json`. This smaller finite example also remains a normalized regression certificate, not a computed certificate for the arbitrarily-small-density theorem.